Use Case Diagram Description: Closet App

Overview

The diagram shows how people and outside services interact with Closet, an intelligent digital wardrobe app. The large rectangle is the system boundary. Everything inside it is something the app does. Everything outside it is an actor that uses the app or supports it.

Two relationships are used. An **«include»** arrow points from a use case to a step it performs every time it runs. An **«extend»** arrow points from an optional behaviour back to the use case it adds to; the user may or may not use it.

Actors

The **User** is the main actor and has a direct line to each of the 14 main use cases. Four external services support the system from outside:

- **Vision Service** suggests a piece's details from its photo.
- **LLM Service** interprets what the user types to the AI Closet Agent.
- **Weather Service** supplies current weather and forecasts.
- **Calendar Service** supplies the user's upcoming events.

The virtual try-on shows a stylised avatar drawn inside the app, and reminders are in-app notifications, so neither needs an external actor.

Main use cases

Register / Login lets the user create an account and sign in.

Manage Style Profile holds the user's taste: styles, colours, occasions, dress code and the modest option. The first time, it is extended by **Complete Style Quiz**.

Add Clothing Item always includes **Upload Photo**, so every piece enters the closet as a photo with details. **AI Pre-fill Details**, supported by the Vision Service, is an optional extension: it suggests category, colour, fabric and dress code, and the user confirms them before anything is saved.

Edit Clothing Item lets the user change a piece's details later.

Delete Clothing Item always includes **Record Deletion Reason** (became small, torn or damaged, no longer my style, or donated, sold or other), so the app learns why an item left the closet.

Manage Availability & Laundry sets each item's status: Available, Laundry, Borrowed, Packed or Unavailable. The laundry basket is the list of items in Laundry, and **Mark All as Clean** is an optional extension that returns them all at once. Only Available items are ever used in outfits.

Chat with AI Closet Agent lets the user talk to the app about their closet. It always includes **Interpret Request**, supported by the LLM Service, and **Analyze Prefs & Feedback**, which draws on favorites, rejections and ratings.

Style This / Forgotten Piece lets the user pick one item and get outfits built around it. It includes **Get Outfit Recommendation**.

Get Outfit Recommendation is the core feature. Every time it runs, it includes:

- **Analyze Prefs & Feedback**
- **Check Item Availability**, which removes every item that is not Available
- **Get Weather**, supported by the Weather Service
- **Generate & Explain 3 Outfits**

It then offers optional extensions, which the user may or may not use: **Virtual Try-On (avatar)**, **Save Outfit**, **Change One Piece** and **Reject Outfit**. **Favorite Outfit** extends Save Outfit, because a favorite is a saved outfit marked with a star. Rejecting can be extended further with **Give Rejection Reason** and **Temporarily Hide Item**.

Plan Outfit Ahead prepares an outfit for a future event. It includes **Read Calendar Event**, supported by the Calendar Service, and **Get Outfit Recommendation**. **Receive Outfit Reminder** is an optional extension, delivered as an in-app notification.

Build Outfit lets the user pick their own pieces. It includes **Check Outfit Compatibility**, which checks completeness, colour harmony, dress code, weather and style.

Mark Outfit as Worn records that the user wore an outfit. It includes **Update Wear History**, and **Rate Outfit After Wearing** is optional.

View Wear Analytics shows how the user wears their wardrobe. Optional extensions are **View Most Worn Items**, **View Least Worn Items**, **View Forgotten Pieces** and **View Utilization Score**.

View Smart Shopping List suggests what to buy. It includes **Analyze Wardrobe Gaps**, **Check Similar Owned Items**, so the app doesn’t suggest something the user already has, and **Read Deletion Reasons**, so a torn or too-small item is suggested as a replacement and an item removed as “no longer my style” is never suggested again.